HW 2 HE Yanning 何衍宁 12311512

1.

1.
A silicon sample is doped with donor impurities to a concentration of 1×10^{15} cm^{-3}. Describe how the electron concentration in the conduction band changes with temperature. You may use the concentration vs temperature plot to explain.

$N_D = 10^{15}\ cm^{-3}$

Freeze-out region

As temperature increase, electron concentration rises sharply as donor ionize, then keep at ≈ $10^{15}\ cm^{-3}$, and finally increase again exponentially in intrinsic region.

Extrinsic region

2.

2.
Consider the two energy levels E_1 and E_2 with $E_1 > E_2$ and an energy separation of 1.12 eV. Assume that the Fermi level EF is in between the two levels and that T = 300 K. If $E_1 - E_F = 0.30$ eV, determine the probability that an energy state at $E = E_1$ is occupied by an electron and the probability that an energy state at $E = E_2$ is empty.

[9.2×10^{-6}, 1.73×10^{-14}]

$E_1 - E_2 = 1.12eV$

$E_1 - E_F = 0.30eV$

$T = 300K$

$k_B = 1.38\times10^{-23}\ J/K$

$$f(E) = \frac{1}{1+\exp\left[\frac{E-E_F}{k_BT}\right]}$$

$$f(E_1) = \frac{1}{1+\exp\left[\frac{E_1-E_F}{k_BT}\right]}$$

$$\Rightarrow f(E_1) \approx 9.2\times10^{-6}$$

$$E_2 - E_F = -0.82eV$$

$$f(E_2) = \frac{1}{1+\exp\left[\frac{E_2-E_F}{k_BT}\right]}$$

$$\Rightarrow f(E_2) = 1.73\times10^{-14}$$

3.

3.

Consider a silicon crystal doped with boron atoms to a concentration of 5×10^{17} cm^{-3} at 300K,

a. Determine the majority and minority carrier concentrations,

b. Determine the position of the Fermi energy level inside the bandgap.

Take n_i to be 1.5×10^{10} cm^{-3}.

[5×10^{17} cm^{-3}, 450 cm^{-3}, 0.448 eV below Ei]

a) $T = 300K$, $N_A = 5\times10^{17}\ cm^{-3}$, $n_i = 1.5\times10^{10}\ cm^{3}$

P-type ⇒ major: holes, minor: electrons

↳ majority: $p \approx N_A = 5\times10^{17}\ cm^{-3}$

↳ minority: $n = \frac{n_i^2}{p} = 4.5\times10^{2}\ cm^{-3} = 450\ cm^{-3}$

b) Fermi energy level

$E_F - E_i = -k_B \cdot T \cdot \ln\left(\frac{p}{n_i}\right)$

$\Rightarrow E_F - E_i \approx -0.448eV$

position: 0.448eV below E_i

4.

4.

A pure silicon sample maintained at room temperature has an intrinsic carrier concentration of 1.5×10^{10} cm^{-3}. It is first doped with donors of concentration 2×10^{14} cm^{-3}, followed by acceptors of concentration 4 x 10^{14} cm^{-3}. Assuming that the carrier mobilities are $\mu_n = 1350$ cm^2/Vs and $\mu_p = 480$ cm^2/Vs,

a. Calculate the majority and minority carrier concentrations,

b. What is the resistivity of the pure sample, prior to the two types of dopings?

c. How will the resistivity change after the dopings?

[2×10^{14} cm^{-3}, 1.125×10^{6} cm^{-3}, 2×10^{5} Ω-cm, 65 Ω-cm]

$T = 300K$, $n_i = 1.5\times10^{10}\ cm^{-3}$

$N_D = 2\times10^{14}\ cm^{-3}$, $N_A = 4\times10^{14}\ cm^{-3}$

$\mu_n = 1350\ cm^2/Vs$, $\mu_p = 480\ cm^2/Vs$

a) net dope: $N_A - N_D = 4\times10^{14} - 2\times10^{14} = 2\times10^{14}\ cm^{-3}$

⇒ p-type: major. holes; minor. electrons major: $p \approx N_A - N_D = 2\times10^{14}\ cm^{-3}$

minor: $n = \frac{n_i^2}{p} = 1.125\times10^{6}\ cm^{-3}$

b) pure: $n = p = n_i = 1.5\times10^{10}\ cm^{-3}$

$\sigma_i = q\cdot n_i\cdot(\mu_n+\mu_p)$

$\Rightarrow \sigma_i = 4.392\times10^{-6}\ S/cm$

$\hookrightarrow \rho_i = \frac{1}{\sigma_i} \approx 2.277\times10^{5}\ \Omega\cdot cm \approx 2\times10^{5}\ \Omega\cdot cm$

c) after doping

major. holes $\Rightarrow p = 2\times10^{14}\ cm^{-3}$

$\hookrightarrow \sigma \approx q\cdot p\cdot\mu_p = 1.536\times10^{-2}\ S/cm$

$\hookrightarrow \rho = \frac{1}{\sigma} \approx 65\ \Omega\cdot cm$

5

Consider a homogenous gallium arsenide semiconductor at T = 300 K with donor concentration of 10^{16} cm^{-3} and the acceptor concentration of 0. *(a)* Calculate the thermal-equilibrium values of electron and hole concentrations. *(b)* For an applied electric field of 10 V/cm, calculate the drift current density. *(c)* Repeat parts *(a)* and *(b)* if the donor concentration is 0 and the acceptor concentration is 10^{16} cm^{-3}.
The intrinsic carrier concentration of GaAs at 300K is 1.8×10^{6} cm^{-3}. Refer to the below mobility Vs. carrier concentrations plot from the text book for the mobility of the GaAs at particular doping.

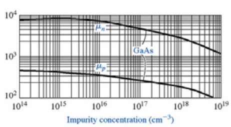

Fig. Q5, the electron and hole mobility of GaAs after doping with different concentrations of donor and acceptor impurities, respectively.

[10^{16} cm^{-3}, 3.24×10^{-4} cm^{-3}, 120A/cm^2, 3.24×10^{-4} cm^{-3}, 10^{16} cm^{-3}, 4.96 A/cm^2]

$T = 300K$, $n_i = 1.8\times10^{-6}\ cm^{-3}$

a) $N_D = 10^{16}\ cm^{-3}$, $N_A = 0$

$n \approx N_D = 10^{16}\ cm^{-3}$

$p = \frac{n_i^2}{n} = 3.24\times10^{-4}\ cm^{-3}$

n-type : major , electrons ; minor , holes

electron : $10^{16}\ cm^{-3}$

hole : $3.24 \times 10^{-4}\ cm^{-3}$

b) $J = q(n \cdot \mu_n + p \cdot \mu_p)$, $E = 10\ V/cm$

n-type : $J \simeq q \cdot n \cdot \mu_n \cdot E = 120\ A/cm^2$

c) $N_D = 0$, $N_A = 10^{16}\ cm^{-3}$

$\Rightarrow$ a p-type : major, hole ; minor . electron

hole : $p \simeq N_A = 10^{16}\ cm^{-3}$

electron : $n = \frac{n_i^2}{p} = 3.24 \times 10^{-4}\ cm^{-3}$

$\Rightarrow$ b $J \simeq q \cdot p \cdot \mu_p \cdot E = 4.96\ A/cm^2$

6.

6.

a. Consider the conductivity of a semiconductor, $\sigma = en\mu_e + ep\mu_h$. Will doping always increase the conductivity?

b. Show that the minimum conductivity for Si is obtained when it is *p*-type doped such that the hole concentration is

$$p_m = n_i \sqrt{\frac{\mu_e}{\mu_h}}$$

and the corresponding minimum conductivity (maximum resistivity) is

$$\sigma_{min} = 2en_i\sqrt{\mu_e\mu_h}$$

c. Calculate p_m and σ_{min} for Si and compare with intrinsic values Use $\mu_n = 1350$ cm²/Vs, $\mu_p = 450$ cm²/Vs, $n_i = 1.45 \times 10^{10}$ cm^{-3} for Si.

[2.51×10^{10} cm^{-3}, 3.62×10^{-6} Scm^{-1}]

a) $\sigma = en_i\mu_n + ep\mu_p$

Doping will always increase the conductivity.

since doping can increase carrier concentration.

b) $\sigma = e(n \cdot \mu_e + p \cdot \mu_h)$, $n = \frac{n_i^2}{p} \Rightarrow \sigma = e\left(\frac{n_i^2}{p}\mu_e + p \cdot \mu_h\right)$

$$\frac{d\sigma}{dp} = e\left(-\frac{n_i^2}{p^2} \cdot \mu_e + \mu_h\right) \Rightarrow \frac{d\sigma}{dp} = 0$$

$$\Rightarrow \mu_h = \frac{n_i^2}{p^2}\mu_e$$

$$\Rightarrow p_m = n_i\sqrt{\frac{\mu_e}{\mu_h}}$$

electron concentration: $n_m = \frac{n_i^2}{p_m}$

$$\sigma_{min} = e(n_m \mu_e + p_m \cdot \mu_h)$$

$$\Rightarrow \sigma_{min} = 2en_i\sqrt{\mu_e \cdot \mu_h}$$

c) $n_i = 1.45 \times 10^{10}\,cm^{-3}$, $\mu_n = 1350\,cm^2/Vs$, $\mu_p = 450\,cm^2/Vs$

$$p_m = n_i\sqrt{\frac{\mu_n}{\mu_p}} \approx 2.51 \times 10^{10}\,cm^{-3}$$

$$\sigma_{min} = 2en_i \cdot \sqrt{\mu_n \cdot \mu_p} \approx 3.62 \times 10^{-6}\,S/cm$$

7.

a. A Si wafer has been doped *n*-type with 10^{17} As atoms cm^{-3}.
1. Calculate the conductivity of the sample at 27 °C.
2. Where is the Fermi level in this sample at 27 °C with respect to the Fermi level (E_{Fi}) in intrinsic Si?
3. Calculate the conductivity of the sample at 127 °C.

b. The above *n*-type Si sample is further doped with 9×10^{16} boron atoms (*p*-type dopant) per centimetre cubed.
1. Calculate the conductivity of the sample at 27 °C.
2. Where is the Fermi level in this sample with respect to the Fermi level in the sample in (a) at 27 °C? Is this an *n*-type or *p*-type Si?

Refer to the below graphs for the parameters required for the calculation.

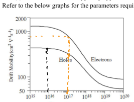

Fig. Q7a. The variation of the drift mobility with dopant concentration in Si for electron and holes at 300 K.

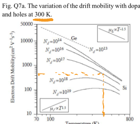

Fig. Q7b. The plot for drift mobility versus temperature for *n*-type Ge and *n*-type Si samples. Various donor concentrations for Si are shown, N_d are in cm^{-3}.

[12.8 Scm^{-1}, 0.407 eV above E_i, 7.21 Scm^{-1}, 1.12 Scm^{-1}, 0.348 eV above E_i]

8.

$N_D = 10^{17}\,cm^{-3}$

$27°C = 300\,K$

$127°C = 400\,K$

$N_A = 9 \times 10^{16}\,cm^{-3}$

$n_i = 1.45 \times 10^{10}\,cm^{-3}$

a) i. $n \approx N_D = 10^{17}\,cm^{-3}$

$p = \frac{n_i^2}{n} \approx 2.1 \times 10^3\,cm^{-3}$ From figure $\mu_n \approx 800\,cm^2/Vs$

$\Rightarrow \sigma = en \cdot \mu_n = 12.8\,S/cm$

ii. n-type $E_F - E_i = kT \cdot \ln\left(\frac{n}{n_i}\right)$

$\Rightarrow E_F - E_i = 0.407\,eV \Rightarrow 0.407\,eV$ above E_i

iii. $T = 400K$ From figure $\mu_n = 450\,cm^2/Vs$

$\sigma = en \cdot \mu_n = 7.21\,S/cm$

b) i. $N_{net} = N_D - N_A = 10^{16}\,cm^{-3}$

$n \approx N_{net} = 10^{16}\,cm^{-3} \Leftarrow \mu_n = 700\,cm^2/Vs$

$\sigma = en \cdot \mu_n = 1.12\,S/cm$

ii. $n_a = 10^{17}\,cm^{-3}$, $n_b = 10^{16}\,cm^{-3}$

$\Delta E_F = kT \cdot \ln\frac{n_a}{n_b} \approx 0.0596\,eV$

$E_{Fb} = E_{Fa} - \Delta E_F = 0.348\,eV \Rightarrow 0.348\,eV$ above E_i

8. In a particular semiconductor material, $\mu_n = 1000$ cm²/V-s, $\mu_p = 600$ cm²/V-s, and $N_C = N_V = 10^{19}$ cm^{-3}. These parameters are independent of temperature. The measured conductivity of the intrinsic material is $\sigma = 10^{-6}$ (Ω-cm)$^{-1}$ at $T = 300$ K. Find the conductivity at $T = 500$ K.

[5.81×10^{-3} Scm^{-1}]

$\sigma(300K) = 10^{-6}\,S/cm$

$n_i = \sqrt{N_C N_V} \cdot \exp\left(-\frac{E_g}{2kT}\right)$, $\sigma = q n_i (\mu_n + \mu_p)$

$\Rightarrow \sigma_0 = q(\mu_n + \mu_p)\sqrt{N_C N_V} \cdot \exp\left(-\frac{E_g}{2kT}\right)$

$\hookrightarrow \sigma(300K) = 10^{-6}\,S/cm$

$\Rightarrow E_g \approx 1.12\,eV$

$$n_i(T) = \sqrt{N_c N_v} \cdot \exp\left(-\frac{E_g}{2kT}\right)$$

$\Rightarrow n_i(500K) \approx 2.29 \times 10^{13}\,cm^{-3}$

$\Rightarrow \sigma(500K) = q n_i(500K)(\mu_n + \mu_p) \approx 5.81 \times 10^{-3}\,S/cm$

9.

9.

Show that the intrinsic carrier concentration is related to the energy bandgap as

$$n_i = \sqrt{N_c N_v} exp\left[\frac{-E_g}{2k_B T}\right]$$

Based on the material parameters given in the following table, calculate the intrinsic carrier concentrations of silicon and germanium at 300K.

Table 1:

	Silicon	Germanium
N_c[cm^{-3}]	2.8×10^{19}	1.04×10^{19}
N_v[cm^{-3}]	1.04×10^{19}	6.0×10^{18}
Bandgap[eV]	1.12	0.66

[6.8×10^{9} cm^{-3}, 2.3×10^{13} cm^{-3}]

i. Show

$$n = N_c \cdot \exp\left(-\frac{E_c - E_F}{kT}\right),\quad p = N_v \cdot \exp\left(-\frac{E_F - E_v}{kT}\right)$$

intrinsic : $n = p = n_i$

$\hookrightarrow E_c - E_F = \frac{E_g}{2},\quad E_F - E_v = \frac{E_g}{2}$

$$\Rightarrow n_i^2 = N_c N_v \cdot \exp\left(-\frac{E_g}{kT}\right)$$

$$\Rightarrow n_i = \sqrt{N_c N_v} \cdot \exp\left(-\frac{E_g}{2kT}\right)$$

ii. Si :

$$n_i = \sqrt{(2.8\times10^{19})(1.04\times10^{19})} \cdot \exp\left(-\frac{1.12}{2\cdot kT}\right) = 6.8\times10^{9}\,cm^{-3}$$

Ge :

$$n_i = \sqrt{(1.04\times10^{19})(6.0\times10^{18})} \cdot \exp\left(-\frac{0.66}{2\cdot kT}\right) = 2.3\times10^{13}\,cm^{-3}$$

10.

10.

Three scattering mechanisms are present in a particular semiconductor material. If only the first scattering mechanism were present, the mobility would be $\mu_1 = 2000$ cm^2/V-s, if only the second mechanism were present, the mobility would be $\mu_2 = 1500$ cm^2/V-s, and if only the third mechanism were present, the mobility would be $\mu_3 = 500$ cm^2/V-s. What is the net mobility?

[316 cm^2/V-s]

$$\frac{1}{\mu_{net}} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3}$$

$\hookrightarrow \mu_{net} = \underline{316 \ cm^2/Vs}$